

## BookPilot Sample: Bayesian Inference

Bayesian inference updates a prior probability distribution using a likelihood function derived from observed data.

The result is a posterior over parameters, which can be used for prediction, model comparison, and experimental design.

A conjugate prior makes the posterior analytically tractable.

When conjugacy fails, Markov chain Monte Carlo (MCMC) approximates the posterior by drawing dependent samples.

## Evidence and Model Comparison

The marginal likelihood, also called the evidence, is the integral of likelihood times prior over parameter space.

Bayes factors compare two models by the ratio of evidences.

Overfitting is penalized because broad priors spread mass thinly, lowering the evidence of overly flexible models.

Information criteria such as WAIC approximate this penalty from posterior samples without computing the integral.

## Practical Workflow

1. Specify a generative model and a prior.
2. Collect data and write the likelihood.
3. Compute or sample the posterior.
4. Check posterior predictive simulations.
5. Compare models and refine the story.

Confusing point: a p-value is not a posterior probability that the null hypothesis is true. They answer different questions.